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FORM APPROVED
OMB NO. 0704-0188

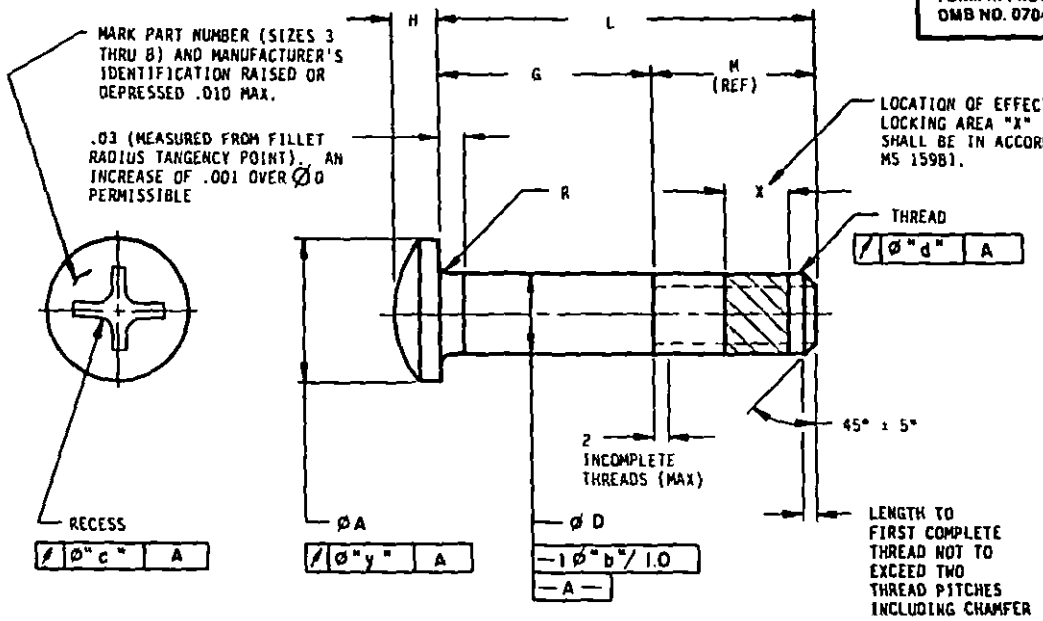



TABLE I. DIMENSIONS AND MECHANICAL PROPERTIES.

| Ø DASH NO | THREAD | ØD MAX / MIN | Ø A MAX / MIN | H MAX / MIN | M MIN REF | R RAD MAX / MIN | "y" | "d" | "c" | "b" | ULTIMATE TENSILE LBS MIN | DOUBLE SHEAR LBS MIN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 04 | .112-40UNC-3A | .1120 / .1095 | .219 / .205 | .080 / .070 | .250 | .010 / .015 | .005 | .0045 | .016 | .0040 | 480 | 930 |
| 06 | .138-32UNC-3A | .1380 / .1355 | .270 / .256 | .097 / .087 | .312 | | | | | | 720 | 1,420 |
| 08 | .164-32UNC-3A | .1635 / .1610 | .322 / .306 | .115 / .105 | .437 | .020 / .010 | | | | | 1,120 | 2,030 |
| 3 | .190-32UNF-3A | .1895 / .1870 | .373 / .357 | .133 / .122 | .469 | | | | | | 1,600 | 2,720 |
| 4 | .250-28UNF-3A | .2495 / .2470 | .492 / .473 | .175 / .162 | .506 | | .006 | | | | 2,910 | 4,700 |
| 5 | .3125-24UNF-3A | .3120 / .3095 | .615 / .594 | .218 / .203 | .531 | | .008 | | | .0030 | 4,640 | 7,370 |
| 6 | .375-24UNF-3A | .3745 / .3720 | .740 / .716 | .261 / .244 | .641 | .025 / .015 | .009 | .0060 | .024 | | 7,020 | 10,600 |
| 7 | .4375-20UNF-3A | .4370 / .4345 | .863 / .838 | .304 / .286 | .656 | | .010 | | | .0025 | 9,490 | 14,400 |
| 8 | .500-20UNF-3A | .4995 / .4970 | .987 / .958 | .348 / .327 | .781 | .030 / .020 | | | | .0020 | 12,790 | 18,800 |

(E) DENOTES CHANGE. SEE PAGE 3 OF 3

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY — AS
AIR FORCE — 99 DLA — IS
REVIEW:
USER:
PROJECT NUMBER: 5306-1648

**MILITARY SPECIFICATION SHEET**

TITLE
BOLT, SELF-LOCKING, 250° F., CRES, 48 KSI Fsu, 80 KSI Ftu, PAN HEAD, CROSS RECESSED

SPECIFICATION SHEET NUMBER
MS21097 20 JAN 92 REV E

SUPERSEDING
MS21097 D 28 JUN 91

AMSC — N/A FSC 5306

DISTRIBUTION STATEMENT
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FORM APPROVED
OMB NO. 0704-0188

TABLE II. GRIP LENGTHS AND LENGTHS.

| LENGTH DASH NO.(a) | GRIP G ±.010 | LENGTH + .031 - .015 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 4-40 | 6-32 | 8-32 | 10-32 | 1/4-28 | 5/16-24 | 3/8-24 | 7/16-20 | 1/2-20 |
| 001 | .062 | .312 | .375 | .500 | | | | | | |
| 002 | .125 | .375 | .437 | .562 | | | | | | |
| 003 | .187 | .437 | .500 | .625 | .656 | | | | | |
| 004 | .250 | .500 | .562 | .687 | .719 | .756 | | | | |
| 006 | .375 | .625 | .687 | .812 | .844 | .881 | .906 | 1.016 | | |
| 008 | .500 | .750 | .812 | .937 | .969 | 1.006 | 1.031 | 1.141 | 1.156 | 1.281 |
| 010 | .625 | .875 | .937 | 1.062 | 1.094 | 1.131 | 1.156 | 1.266 | 1.281 | 1.406 |
| 012 | .750 | 1.000 | 1.062 | 1.187 | 1.219 | 1.256 | 1.281 | 1.391 | 1.406 | 1.531 |
| 014 | .875 | 1.125 | 1.187 | 1.312 | 1.344 | 1.381 | 1.406 | 1.516 | 1.531 | 1.656 |
| 016 | 1.000 | 1.250 | 1.312 | 1.437 | 1.469 | 1.506 | 1.531 | 1.641 | 1.656 | 1.781 |
| 018 | 1.125 | 1.375 | 1.437 | 1.562 | 1.594 | 1.631 | 1.656 | 1.766 | 1.781 | 1.906 |
| 020 | 1.250 | 1.500 | 1.562 | 1.687 | 1.719 | 1.756 | 1.781 | 1.891 | 1.906 | 2.031 |
| 022 | 1.375 | 1.625 | 1.687 | 1.812 | 1.844 | 1.881 | 1.906 | 2.016 | 2.031 | 2.156 |
| 024 | 1.500 | 1.750 | 1.812 | 1.937 | 1.969 | 2.006 | 2.031 | 2.141 | 2.156 | 2.281 |
| 026 | 1.652 | 1.875 | 1.937 | 2.062 | 2.094 | 2.131 | 2.156 | 2.266 | 2.281 | 2.406 |
| 028 | 1.750 | 2.000 | 2.062 | 2.187 | 2.219 | 2.256 | 2.281 | 2.391 | 2.406 | 2.531 |
| 030 | 1.875 | | 2.187 | 2.312 | 2.344 | 2.381 | 2.406 | 2.516 | 2.531 | 2.656 |
| 032 | 2.000 | | 2.312 | 2.437 | 2.469 | 2.506 | 2.531 | 2.641 | 2.656 | 2.781 |
| 034 | 2.125 | | 2.437 | 2.562 | 2.594 | 2.631 | 2.656 | 2.766 | 2.781 | 2.906 |
| 036 | 2.250 | | 2.562 | 2.687 | 2.719 | 2.756 | 2.781 | 2.891 | 2.906 | 3.031 |
| 038 | 2.375 | | 2.687 | 2.812 | 2.844 | 2.881 | 2.906 | 3.016 | 3.031 | 3.156 |
| 040 | 2.500 | | 2.812 | 2.937 | 2.969 | 3.006 | 3.031 | 3.141 | 3.156 | 3.281 |
| 042 | 2.625 | | | | 3.094 | 3.131 | 3.156 | 3.266 | 3.281 | 3.406 |
| 044 | 2.750 | | | | 3.219 | 3.256 | 3.281 | 3.391 | 3.406 | 3.531 |
| 046 | 2.875 | | | | 3.344 | 3.381 | 3.406 | 3.516 | 3.531 | 3.656 |
| 048 | 3.000 | | | | 3.469 | 3.506 | 3.531 | 3.641 | 3.656 | 3.781 |
| 050 | 3.125 | | | | 3.594 | 3.631 | 3.656 | 3.766 | 3.781 | 3.906 |
| 052 | 3.250 | | | | | | | | | 4.031 |
| 054 | 3.375 | | | | | | | | | 4.156 |
| 056 | 3.500 | | | | | | | | | 4.281 |
| 058 | 3.625 | | | | | | | | | 4.406 |
| 060 | 3.750 | | | | | | | | | 4.531 |
| 062 | 3.875 | | | | | | | | | 4.656 |
| 064 | 4.000 | | | | | | | | | 4.781 |
| 066 | 4.125 | | | | | | | | | 4.906 |

(a) UNASSIGNED DASH NUMBERS SHALL NOT BE USED EXCEPT THOSE DASH NUMBERS INCLUDED BY THE FOLLOWING CODING NOTES: DASH NUMBERS INDICATE PREFERRED GRIP LENGTHS IN .125 INCREMENTS FOR LENGTHS LONGER THAN .250 INCHES. INTERMEDIATE GRIP BETWEEN THOSE LENGTHS LISTED MAY BE SPECIFIED IN .062 INCREMENTS BY INTERMEDIATE DASH NUMBERS. LENGTH AND GRIP OF A BOLT WITH AN ODD NUMBER FOR THE DASH NUMBER WILL BE .062 GREATER THAN SHOWN FOR THE BOLT WITH THE PRECEDING EVEN DASH NUMBER.

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY — AS
AIR FORCE — 99 DLA — IS
REVIEW:
USER:
PROJECT NUMBER: 5306-1648

MILITARY SPECIFICATION SHEET

TITLE
BOLT, SELF-LOCKING, 250° F, CRES, 48 KSI Fsu, 80 KSI Ftu, PAN HEAD, CROSS RECESSED

SPECIFICATION SHEET NUMBER
MS21097 20 JAN 92 REV E

SUPERSEDING
MS21097 D 28 JUN 91

AMSC — N/A FSC 5306

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FORM APPROVED
OMB NO. 0704-0188

REQUIREMENTS:

1. MATERIAL: CORROSION RESISTANT STEEL PER QQ-S-763 THAT WILL PROVIDE IN THE FINISHED BOLT THE STRENGTH AND PERMEABILITY SPECIFIED.
2. TREATMENT: PASSIVATED PER QQ-P-35.
3. SURFACE TEXTURE: THE SURFACE OF THE SHANK, FILLET AND BEARING AREA SHALL NOT EXCEED 125 MICROINCHES PER ANSI/ASME B46.1.
4. MAGNETIC PERMEABILITY: THESE BOLTS HAVE A MAGNETIC PERMEABILITY OF 2.0 MAXIMUM (AIR =1.0) FOR A FIELD OF H=200 OERSTEDS USING MAGNETIC INDICATOR PER MIL-I-17214.
5. LOCKING ELEMENT: IN ACCORDANCE WITH MIL-F-18240. LOCKING ELEMENT AREA, WITH LOCKING ELEMENT INCORPORATED, MUST PASS A RING GAGE EQUAL TO THE MAXIMUM MAJOR THREAD DIAMETER +.0035, -.0000 FREELY OR WITH FINGER PRESSURE.
6. RECESS: IN ACCORDANCE WITH MS9006.
7. THREAD FORMING: THE THREADS SHALL BE FORMED BY ANY SINGLE ROLLING PROCESS AND IN ACCORDANCE WITH MIL-S-7742.
8. ULTIMATE TENSILE STRENGTH: BASED ON 80 KSI ULTIMATE TENSILE STRENGTH AND CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2.

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. DIMENSIONS AND TOLERANCING SHALL BE INTERPRETED IN ACCORDANCE WITH ANSI/ASME Y14.5M.
3. FOR DESIGN AND USAGE LIMITATIONS SEE MS15981.
4. GRIP LENGTH "G" IS MEASURED FROM UNDERSIDE OF HEAD TO THE END OF FULL CYLINDRICAL PORTION OF SHANK " D".
5. REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION OR MANUFACTURING REQUIREMENT.
6. EXAMPLE OF PART NUMBER: MS21097-4008 = .2500-28 BOLT, .500 GRIP.
7. IN EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
8. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THIS ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN. H
9. EXISTING INVENTORY, I.E. WITHOUT MANUFACTURER'S IDENTIFICATION AND PART NUMBER, MAY BE USED UNTIL EXAUSTED. (E)

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AY
NAVY — AS
AIR FORCE — 99
DLA — IS
REVIEW:
USER:
PROJECT NUMBER: 5306-1648

MILITARY SPECIFICATION SHEET

TITLE
BOLT, SELF-LOCKING, 250° F.
CRES, 48 KSI Fsu, 80 KSI Ftu.
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